
Examiners' commentaries 2024 (May)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2023–24. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

General remarks

Learning outcomes

At the end of this half course and having completed the essential reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

Key steps to improvement

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2** and **ST1215 Introduction to mathematical statistics**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2**, **ST1215 Introduction to mathematical statistics** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only

on the past papers. Candidates should practise on all the examples, activities and sample examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = (f_X(x; \theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$\begin{aligned} l_{\mathbf{X}}(\theta; \mathbf{x}) &= \log(f_{\mathbf{X}}(\theta; \mathbf{x})) \\ &= \log \left(\prod_{i=1}^n f_{X_i}(x_i; \theta) \right) \\ &= \sum_{i=1}^n \log f_{X_i}(x_i; \theta). \end{aligned}$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This may be due to a number of reasons, but one particular failing is ‘**question spotting**’, that is, confining your examination preparation to a few questions and/or topics which have come up in past papers for the course. This can have serious consequences.

We recognise that candidates might not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet available on the VLE. You should read the syllabus carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers. Examination papers may legitimately include questions on any topic in the syllabus. So, although past papers can be helpful during your revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question-spotting strategy, it is likely you will find yourself in difficulties when you sit the examination. We strongly advise you not to adopt this strategy.

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Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

- (a) Let $X = \{X_1, X_2, \dots, X_n\}$ be a random sample from a smooth probability density function $f_X(x; \theta)$, where θ is an unknown parameter. Let $s_X(\theta; X)$ denote the score function. Show that $E(s_X(\theta; X)) = 0$.

(You may assume sufficient regularity conditions to allow differentiation under integral signs.)

(10 marks)

Reading for this question

Section 2.10.2 of the subject guide presents the proof of the expectation of the score function.

Approaching the question

For the continuous case:

$$\begin{aligned}
 E(s_{\mathbf{X}}(\theta; \mathbf{X})) &= \int_{\mathbb{R}^n} s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\
 &= \int_{\mathbb{R}^n} \frac{1}{L_{\mathbf{X}}(\theta; \mathbf{x})} \frac{\partial}{\partial \theta} L_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\
 &= \int_{\mathbb{R}^n} \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} && (\text{since } L_{\mathbf{X}}(\theta; \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}; \theta)) \\
 &= \frac{\partial}{\partial \theta} \underbrace{\int_{\mathbb{R}^n} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x}}_{=1} \\
 &= \frac{d}{d\theta} 1 \\
 &= 0
 \end{aligned}$$

because the last integral is equal to one, since the density function is normalised.

This was a standard bookwork question which was straightforward for candidates who had studied it.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a normal distribution with mean μ and variance σ^2 . Consider the sample mean and the sample variance. It is known that in this case:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \quad \text{where} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

and where χ_{n-1}^2 denotes a chi-squared distribution with $n-1$ degrees of freedom. It is also known that if $Y \sim \chi_k^2$, then $E(Y) = k$ and $\text{Var}(Y) = 2k$.

- i. Find the mean squared error of S^2 as an estimator of σ^2 . (6 marks)
- ii. Consider the alternative estimator of σ^2 given by $(n-1)S^2/n$. Find its mean squared error and compare this with the mean squared error of the estimator S^2 . (6 marks)

Reading for this question

Section 3.4.1 of the subject guide presents the mean squared error, while Example 3.8 is highly relevant for this question.

Approaching the question

- i. We have:

$$E\left(\frac{(n-1)S^2}{\sigma^2}\right) = n-1 \quad \Rightarrow \quad \frac{n-1}{\sigma^2} E(S^2) = n-1 \quad \Rightarrow \quad E(S^2) = \sigma^2$$

and:

$$\text{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \quad \Rightarrow \quad \frac{(n-1)^2}{\sigma^4} \text{Var}(S^2) = 2(n-1) \quad \Rightarrow \quad \text{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

Since $E(S^2) = \sigma^2$, the estimator S^2 is unbiased. The mean squared error of S^2 is:

$$\text{MSE}(S^2) = \text{Var}(S^2) + (\text{Bias}(S^2))^2 = \frac{2\sigma^4}{n-1}.$$

ii. The bias of the estimator $(n-1)S^2/n$ is given by:

$$\text{Bias}\left(\frac{n-1}{n} S^2\right) = E\left(\frac{n-1}{n} S^2\right) - \sigma^2 = \frac{n-1}{n} E(S^2) - \sigma^2 = -\frac{\sigma^2}{n}.$$

The variance of the estimator $(n-1)S^2/n$ is given by:

$$\text{Var}\left(\frac{n-1}{n} S^2\right) = \left(\frac{n-1}{n}\right)^2 \text{Var}(S^2) = \frac{(n-1)^2}{n^2} \frac{2\sigma^4}{n-1} = \frac{2(n-1)}{n^2} \sigma^4.$$

Hence the mean squared error of the estimator $(n-1)S^2/n$ is:

$$\text{MSE}(S^2) = \text{Var}(S^2) + (\text{Bias}(S^2))^2 = \frac{2(n-1)}{n^2} \sigma^4 + \frac{\sigma^4}{n^2} = \frac{2n-1}{n^2} \sigma^4.$$

Note that the mean squared error of the estimator $(n-1)S^2/n$ is smaller than that of the estimator S^2 for $n > 2$.

Some candidates used imprecise notation, in particular ' σ^2 ' for one of the estimators, however this should be avoided as σ^2 is the actual parameter for which these are estimators!

(c) **A random sample consists of a single observation of X which follows a binomial distribution, specifically $\text{Bin}(4, \pi)$. Suppose we are interested in testing:**

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi = 0.25.$$

A statistical test rejects H_0 if the observed value x is 0. Calculate the *size* and the *power* of the test.

(8 marks)

Reading for this question

Section 5.4.3 of the subject guide presents the size and power of statistical tests.

Approaching the question

To find the size of the test we need to find the probability given by:

$$P(\text{reject } H_0 \mid \pi = 0.50)$$

and for the power we need to calculate the probability given by:

$$P(\text{reject } H_0 \mid \pi = 0.25).$$

Since we reject H_0 if $x = 0$ and $X \sim \text{Bin}(4, \pi)$, the size of the test is:

$$P(\text{reject } H_0 \mid \pi = 0.50) = P(X = 0 \mid \pi = 0.50) = \binom{4}{0} (1 - 0.50)^4 = \frac{1}{16} = 0.0625$$

and the power of the test is:

$$P(\text{reject } H_0 \mid \pi = 0.25) = P(X = 0 \mid \pi = 0.25) = \binom{4}{0} (1 - 0.25)^4 \approx 0.3164.$$

Some candidates mixed up size and power, conditioning on the wrong hypothesis. There were also some arithmetic mistakes.

(d) i. **Provide the definition of a sufficient statistic.**

(4 marks)

ii. **State the factorisation theorem for the continuous case.**

(6 marks)

Reading for this question

Section 2.8.1 of the subject guide defines sufficient statistics, and Section 2.8.3 states the factorisation theorem.

Approaching the question

- i. Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. A statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a sufficient statistic for $\boldsymbol{\theta}$ if the conditional distribution of $\mathbf{X}$ given $\mathbf{U}$ does not depend on $\boldsymbol{\theta}$.
- ii. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with joint density function $f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$. The statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is sufficient for the vector of parameters $\boldsymbol{\theta}$ if and only if we can find functions g and h such that:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. Note $\mathbf{T}(\mathbf{X})$ and $\boldsymbol{\theta}$ can be scalars, denoted $T(\mathbf{X})$ and θ , respectively.

This was a standard bookwork question which was straightforward for candidates who had studied these fundamental concepts of the course.

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2\}$ be a random sample of size $n = 2$ from an exponential distribution with parameter $\mu > 0$, such that:

$$f_X(x; \mu) = \begin{cases} \mu^{-1} e^{-x/\mu} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find a sufficient statistic for μ .

(5 marks)

- (b) Consider the estimator $\hat{\mu} = X_1$. Show that $\hat{\mu}$ is an unbiased estimator of μ . Explain why it is a poor estimator.

(3 marks)

- (c) Find the conditional expectation of $\hat{\mu}$ given $T = X_1 + X_2$. Confirm that this conditional expectation is a better estimator than $\hat{\mu}$ in terms of the mean squared error.

(Hint: Find the probability density function of T and the joint probability density function of T and X_1 first.)

(12 marks)

Reading for this question

Sections 2.8.1, 3.4.1 and 3.5 of the subject guide present the relevant material for this question.

Approaching the question

- (a) The joint probability density function of the sample is:

$$f_{\mathbf{X}}(x_1, x_2; \mu) = \frac{1}{\mu} e^{-x_1/\mu} \frac{1}{\mu} e^{-x_2/\mu} = \frac{1}{\mu^2} e^{-(x_1+x_2)/\mu}$$

where if we set:

$$g(x_1 + x_2, \mu) = \mu^{-2} \exp\left(-\frac{x_1 + x_2}{\mu}\right) \quad \text{and} \quad h(\mathbf{x}) = 1$$

we have:

$$f_{\mathbf{X}}(x_1, x_2; \mu) = g(x_1 + x_2, \mu) h(\mathbf{x}).$$

Therefore, the statistic $T(X_1, X_2) = X_1 + X_2$ is a sufficient statistic for μ .

- (b) We have $E(X_1) = \mu$, hence X_1 is an unbiased estimator of μ . However, since it is based only on X_1 , it fails to take into account all the information in the sample, i.e. X_2 is not used, hence it is a poor estimator. Also, accept that X_i is not a consistent estimator.
- (c) **Method A:**

The distribution of $T = X_1 + X_2$ is $\text{Gamma}(2, \mu)$. To find the joint distribution of (X_1, T) , we may use the transformation from (X_1, X_2) to (T, X_1) . The Jacobian of the transformation is 1, and so we get:

$$f_{X_1, T}(x_1, t; \mu) = \frac{1}{\mu^2} \exp\left(-\frac{t - x_1 + x_1}{\mu}\right) = \frac{1}{\mu^2} \exp\left(-\frac{t}{\mu}\right)$$

for $0 \leq x_1 \leq t < \infty$. Hence the conditional pdf of X_1 given T is:

$$f_{X_1 | T}(x_1 | T = t) = \frac{f_{X_1, T}(x_1, t; \mu)}{f_T(t; \mu)} = \frac{\mu^{-2} \exp(-t/\mu)}{(\Gamma(2))^{-1} \mu^{-2} t \exp(-t/\mu)} = \frac{1}{t}$$

for $0 \leq x_1 \leq t < \infty$. Hence, conditional on $T = t$, X_1 is $\text{Uniform}[0, t]$, and $E(X_1 | T = t) = t/2$. Therefore, the improved estimator of μ is $T/2$, i.e. the sample mean. We have $E(T/2) = 2\mu/2 = \mu$ and:

$$\text{MSE}\left(\frac{T}{2}\right) = \text{Var}\left(\frac{T}{2}\right) = \frac{\mu^2}{2} < \mu^2 = \text{Var}(X_1) = \text{MSE}(X_1).$$

Therefore, the new estimator is better in terms of the mean squared error.

Method B:

Since X_1 and X_2 are identically distributed:

$$E(X_i | T = t) = E(X_1 | T = t) = E(X_2 | T = t).$$

The Rao–Blackwell theorem states that the following estimator is better:

$$\begin{aligned} \hat{\mu}_2 = E(\hat{\mu}_1 | T = t) &= E(X_1 | T = t) = \frac{1}{2}(E(X_1 | T = t) + E(X_1 | T = t)) \\ &= \frac{1}{2}(E(X_1 | T = t) + E(X_2 | T = t)) \\ &= \frac{1}{2} E(X_1 + X_2 | T = t) \\ &= \frac{1}{2} E(T | T = t) \\ &= \frac{t}{2}. \end{aligned}$$

See Method A for showing why this is better in terms of the mean squared error.

Candidates found part (c) the most challenging part of this question, although note that there was a choice of two approaches, albeit Method A uses a technique based on the complementary course **ST2133 Advanced statistics: distribution theory**, which note is a co-requisite for this course, hence candidates would be expected to have the knowledge to use Method A.

Question 3

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a distribution with the probability density function:

$$f_{\mathbf{X}}(x; \beta) = \begin{cases} \beta x^{\beta-1} & \text{for } 0 < x < 1 \text{ and } \beta > 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine a one-dimensional sufficient statistic for β . (3 marks)
- (b) Obtain the score function and find Fisher's information for β . Also, find the maximum likelihood estimator of β . (8 marks)
- (c) Describe how a maximum likelihood estimator may be used to form an approximate pivotal function for large sample sizes. (4 marks)
- (d) Determine an asymptotic $100(1 - \alpha)\%$ confidence interval for β . (5 marks)

Reading for this question

Sections 2.8.1, 3.9 and 4.6 of the subject guide present the relevant material for this question.

Approaching the question

- (a) The likelihood function is:

$$L_{\mathbf{X}}(\beta; \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}; \beta) = \prod_{i=1}^n \beta x_i^{\beta-1} = \beta^n \left(\prod_{i=1}^n x_i \right)^{\beta-1}.$$

We can write:

$$f_{\mathbf{X}}(\mathbf{x}; \beta) = g\left(\prod_{i=1}^n x_i, \beta\right) h(\mathbf{x})$$

by setting:

$$g\left(\prod_{i=1}^n x_i, \beta\right) = f_{\mathbf{X}}(\mathbf{x}; \beta) \quad \text{and} \quad h(\mathbf{x}) = 1.$$

Hence by the factorisation theorem $\prod_{i=1}^n X_i$ is a one-dimensional sufficient statistic for β .

- (b) The log-likelihood function is:

$$l_{\mathbf{X}}(\beta; \mathbf{x}) = \log L_{\mathbf{X}}(\beta; \mathbf{x}) = n \log(\beta) + (\beta - 1) \sum_{i=1}^n \log(x_i).$$

The score function is:

$$s_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{\partial}{\partial \beta} l_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{n}{\beta} + \sum_{i=1}^n \log(x_i).$$

Fisher's information for β is:

$$\mathcal{I}_{\mathbf{X}}(\beta) = -E\left(\frac{\partial^2}{\partial \beta^2} l_{\mathbf{X}}(\beta; \mathbf{x})\right) = -E\left(-\frac{n}{\beta^2}\right) = \frac{n}{\beta^2}.$$

Setting $s_{\mathbf{X}}(\beta; \mathbf{x}) = 0$ provides:

$$\hat{\beta} = -\frac{n}{\sum_{i=1}^n \log(X_i)}$$

as a candidate maximum likelihood estimator. Since:

$$\frac{\partial^2}{\partial \beta^2} l_{\mathbf{X}}(\beta; \mathbf{x}) = -\frac{n}{\beta^2} < 0$$

we conclude that this is indeed the maximum likelihood estimator of β .

(c) If $\hat{\theta}$ is a maximum likelihood estimator of θ , then under mild regularity conditions:

$$\hat{\theta} \xrightarrow{d} N(\theta, \mathcal{I}_{\mathbf{X}}^{-1}(\theta)).$$

For large n we have:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\hat{\theta} - \theta) \sim N(0, 1)$$

approximately. As $\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\hat{\theta} - \theta)$ converges to a distribution which does not depend on θ , we say that it is asymptotically pivotal for θ .

Since $\hat{\theta}$ is a consistent estimator of θ , the quantity $\mathcal{I}_{\mathbf{X}}(\hat{\theta})$ converges in probability to $\mathcal{I}_{\mathbf{X}}(\theta)$, which is total Fisher information. Therefore, in a second level of approximation, we can write for large sample sizes that:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\hat{\theta})}(\hat{\theta} - \theta) \sim N(0, 1)$$

approximately, and use this function to construct asymptotic confidence intervals for the parameter θ . The approximation improves as $n \rightarrow \infty$.

(d) We use the maximum likelihood estimator $\hat{\beta}$ from part (b). We know that for large sample sizes:

$$\hat{\beta} \overset{approx}{\sim} N\left(\beta, \frac{\beta^2}{n}\right)$$

since $\mathcal{I}_{\mathbf{X}}(\beta) = n/\beta^2$. In an extra level of approximation we may use:

$$\hat{\beta} \overset{approx}{\sim} N\left(\beta, \frac{\hat{\beta}^2}{n}\right).$$

Let $S_{\beta} = \hat{\beta}^2/n$. We can write:

$$\frac{\hat{\beta} - \beta}{\sqrt{S_{\beta}}} \overset{approx}{\sim} N(0, 1).$$

Using the pivotal function technique for constructing confidence intervals, we get the following asymptotic $100(1 - \alpha)\%$ confidence interval:

$$\left[\hat{\beta} - z_{\alpha/2} \sqrt{S_{\beta}}, \hat{\beta} + z_{\alpha/2} \sqrt{S_{\beta}} \right].$$

Candidates generally performed well in this question. Errors mainly related to the calculus step in part (b), as well as incorrect solving of the expectation when deriving Fisher information.

Question 4

- (a) State the Neyman–Pearson lemma, and explain how it may be used to derive the uniformly most powerful test for a simple null hypothesis against a one-sided alternative hypothesis.

(10 marks)

Reading for this question

Section 5.5.1 of the subject guide presents the Neyman–Pearson lemma.

Approaching the question

Consider a test procedure T of size α for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\theta_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\theta_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{ \mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\theta_1; \mathbf{x})}{L_{\mathbf{X}}(\theta_0; \mathbf{x})} > k_{\alpha} \right\}$$

then T is the most powerful test of size α . Here $L_{\mathbf{X}}$ is the likelihood function and k_{α} is a constant dependent on α . Note that $\alpha = P_{\theta_0}(\mathbf{X} \in R_T)$.

Some suitable comment about applicability to a one-sided alternative hypothesis was expected.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the distribution with probability density function:

$$f_X(x; \theta) = \begin{cases} \frac{2x}{\theta} \exp\left(-\frac{x^2}{\theta}\right) & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

- i. Show that the most powerful test for testing the hypotheses:

$$H_0 : \theta = 1 \quad \text{vs.} \quad H_1 : \theta = 2$$

will reject H_0 if and only if $T > k$ for some constant $k > 0$, where:

$$T = \sum_{i=1}^n X_i^2.$$

(7 marks)

- ii. Would the statistical test from part i. be the uniformly most powerful test for testing the following hypotheses?

$$H_0 : \theta = 1 \quad \text{vs.} \quad H_1 : \theta > 1.$$

Briefly justify your answer.

(3 marks)

Reading for this question

Section 5.6 of the subject guide presents likelihood ratio tests.

Approaching the question

- i. The likelihood function may be written as:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{2x_i}{\theta} \exp\left(-\frac{x_i^2}{\theta}\right) = 2^n \theta^{-n} \left(\prod_{i=1}^n x_i\right) \exp\left(-\frac{\sum_{i=1}^n x_i^2}{\theta}\right).$$

For the case of testing:

$$H_0 : \theta = 1 \quad \text{vs.} \quad H_1 : \theta = 2$$

the likelihood ratio is:

$$\frac{L_{\mathbf{X}}(2; \mathbf{x})}{L_{\mathbf{X}}(1; \mathbf{x})} = \frac{2^{-n} \exp\left(\left(-\frac{1}{2}\right) \sum_{i=1}^n x_i^2\right)}{1^{-n} \exp\left(-\sum_{i=1}^n x_i^2\right)} = 2^{-n} \exp\left(\frac{1}{2} \sum_{i=1}^n x_i^2\right).$$

Since the above ratio is increasing in $\sum_{i=1}^n x_i^2$, the Neyman–Pearson lemma entails that the most powerful test rejects H_0 iff $\sum_{i=1}^n x_i^2 > k$, where $k > 0$ is a constant determined by:

$$P\left(\sum_{i=1}^n X_i^2 > k \mid \theta = 1\right) = \alpha.$$

- ii. Note the test does not depend on the value 2 (which is > 1) of the alternative hypothesis of the previous part. Hence the same test would be the uniformly most powerful test for testing H_0 versus the composite alternative $H_1 : \theta > 1$.

Candidates generally performed well in this question, although there were some algebraic errors when simplifying the likelihood ratio.